Kartheek Mukkavilli

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Education

University of Texas at Austin — Austin, TX

B.S. Electrical & Computer Engineering — GPA: 3.95/4.0 — Expected May 2028

Coursework: Embedded Systems · Digital Logic Design · Digital Signal Processing · Discrete Mathematics · Software Design · Circuit Theory · Algorithms · Signals & Systems · Computer Architecture · Operating Systems · Linear Algebra · Differential Equations

Elements of Computing Certificate (CS) · Business Foundations Minor

Technical Skills

Languages: C, C++, Python, Assembly
Embedded/RTOS: Zephyr RTOS, FreeRTOS, nRF Connect SDK, LTE-M, GNSS, LoRa, MQTT, UART, RTT, I2C, SPI, PWM, ESP-NOW, TLS/mTLS
FPGA/Hardware: Xilinx Vivado, Verilog, nRF9160, ESP32, Raspberry Pi, Arduino, MPU-6050, KiCad, Fusion 360, Saleae Logic Analyzer, nrfjprog, AWS IoT Core
DevOps/Tools: Docker, GitLab CI/CD, Git, Linux

Experience

Southwest Research Institute — San Antonio, TX

Student Engineer, Tactical Aerospace Department (Division 16) — *May 2026 – Present*

- Developing FPGA firmware and bootloader for the F-16 SIU (Sensor Integration Unit) in Xilinx Vivado within an ITAR-controlled environment, consolidating divergent legacy codebases into a single validated monorepo with clean version-control history.
- Validated clean-clone build portability across multiple development machines, eliminating environment-specific toolchain drift ahead of full containerization.
- Building a Dockerized build environment and GitLab CI/CD pipelines to automate synthesis, implementation, and bitstream validation, reducing manual build overhead for the firmware team.

National Oilwell Varco — Houston, TX

Embedded Systems Intern — *Jun – Aug 2025*

- Replaced $20K/site legacy PLC/Modbus installations with a custom nRF9160 board under $250 (98% cost reduction), deployed across 50 field units within a $5K R&D budget.
- Engineered C/Zephyr RTOS firmware for LTE-M + GNSS over a 40-node LoRa mesh to AWS IoT Core; wrote a safe-shutdown handshake preventing LTE tower rejection on power cycle.
- Automated TLS certificate provisioning via Python + AT%CMNG, reducing per-unit onboarding from hours to under one minute across all deployed units.
- Led full product lifecycle – KiCad PCB layout, Fusion 360 enclosure, firmware dev, and field validation – delivering production-ready hardware under budget.

Human Centered Robotics Laboratory — UT Austin — Austin, TX

Research Assistant — *Jan 2026 – Present*

- Architected C++ firmware on Raspberry Pi Pico to synchronize 16-DOF bionic hand kinematics from remote glove telemetry, achieving 1ms sensor-to-actuation latency.
- Tuned PID control loops for 10 independent joints, improving tracking from unreliable to consistently stable real-time replication across the full range of hand motion.
- Debugged the sensor-to-actuation pipeline end-to-end using onboard logging to isolate and reduce latency bottlenecks.

Projects

LiteWing — Custom Quadcopter Flight Controller | *C++, FreeRTOS, ESP32, MPU-6050*

- Programmed a 1kHz FreeRTOS PID attitude control loop; measured 1.0ms loop period with 45µs jitter via GPIO toggling and Saleae logic analyzer.
- Complementary filter IMU fusion via I2C + 400Hz PWM motor control; achieved ±2.5° attitude hold over a 60-second flight test; designed a custom ESP-NOW telemetry and command layer.

nrf9160-zephyr-template | *C, Zephyr RTOS, nRF9160, AWS IoT, LoRa*

- Open-sourced a production Zephyr firmware template for the nRF9160: LTE-M bring-up with exponential backoff, GNSS PVT parsing, AWS IoT MQTT over mTLS, RTT/UART logging, and automated TLS provisioning – generalized from NOV production firmware.

pid-motor-sim — PID Control Simulator | *Python*

- 3-DOF DC motor arm simulation with anti-windup PID, derivative-on-measurement, and Ziegler-Nichols auto-tuning across an 80-config gain sweep with rise time, settling time, and overshoot metrics.

kinematic-transformer | *Python, PyTorch*

- Transformer encoder for real-time multi-joint kinematic forecasting; 0.571ms inference latency – demonstrates ML-to-actuation pipeline design for embedded control systems.

Activities

UT IEEE RAS · TSA Co-Founder (3× National Finals) · 4× Gold PVSA Medalist · SAMPADA Certified Violinist, ABRSM Grade 5